TRACK 02 · EchoGesture

EchoGesture

Sign language → live AR captions, fully on-device

Team Name: [CollegeName_TeamName]

College Name: [College Name]

Members Details

[Member 1 · year · branch]

[Member 2 · year · branch]

[Member 3 · year · branch]

[Member 4 · year · branch]

THE PROBLEM

Signed conversation moves fast. Captions have to keep up, and the camera must keep what it sees to itself.

1. **The conversation gap**
   Natural signing is rapid. If translation lags past ~50 ms, turns fall out of sync and the dialogue feels awkward.

2. **The privacy principle**
   Glasses film bystanders. Raw video of people who never consented must never be uploaded to a cloud server.

3. **The exocentric reality**
   A wearer sees signers from the side, moving, in low light and partly occluded, not a tidy front-facing webcam.

THE BUDGET ON A WEARABLE

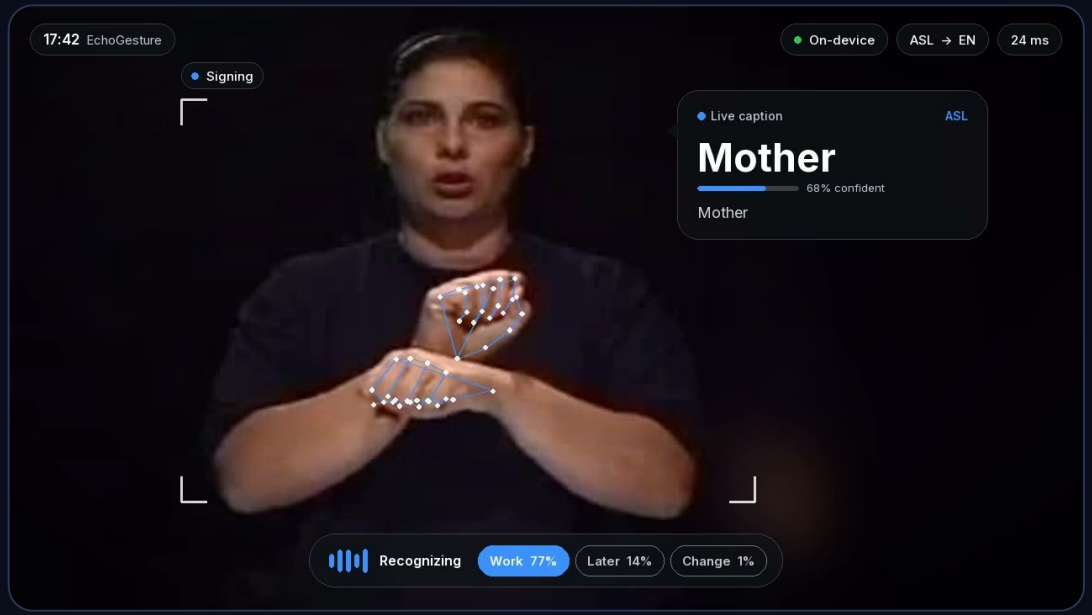
≤ 50 ms
per-frame latency

≤ 20 MB
quantized INT8 model

0 bytes
of video sent to the cloud

2,000
ASL words in WLASL

SOLUTION



Wearer's view: the caption floats beside the signer, never over the face or hands

See what's signed, right where it's signed

- **Keypoints, not pixels**
  MediaPipe tracks 21 points per hand plus the upper body. The model never sees faces or backgrounds.
- **Tiny temporal model**
  A 3-model 1D-CNN ensemble reads 32 frames of motion. INT8, 5.8 MB, 1.6 ms.
- **Glanceable AR caption**
  Words pop in with a confidence meter; unsure guesses show in amber with a “?”.
- **Two-way ready**
  Optional speech output says each word aloud.

ARCHITECTURE

RUNTIME · every stage runs on the device, per frame

Camera frame
Exocentric view
15–30 fps

OpenCV

Keypoints
Hands + upper body
55 3D points

MediaPipe Tasks
hand || pose threads

Features
Body-centred, roll-fixed
hand shape + velocity

32 × 458
occlusion flags

SignNet ×3
Dilated depthwise
temporal CNN

ONNX INT8 · 5.8 MB
1.6 ms

Glass HUD
Anchored caption
+ top-3 + voice

Pillow · OpenCV
macOS say / espeak

TRAINING · offline, once

WLASL100
1,019 clips downloadable
held-out test signers

Augment
mirror · yaw · roll · speed
hand & fingertip dropout

Train
PyTorch · 3 seeds
label smoothing, cosine LR

Quantize
static INT8 QDQ, per-channel
FP32 ↔ INT8 agreement 100%

Python 3.11 | MediaPipe Tasks | PyTorch | ONNX Runtime | OpenCV | Pillow | Docker

TRACK FINAL OUTPUT

results.csv · WLASL100 held-out signers · Mac mini (Apple M1), CPU only

Metric	Target	Measured	Status
Top-1 accuracy	≥ 70 %	**71.4 %**	PASS
Latency p95 / frame	≤ 50 ms	**39.7 ms**	PASS
Model size	≤ 25 MB	**19.4 MB**	PASS
Robustness drop	≤ 5 pt	**3.0 pt**	PASS
Thermal drift, 10 min	≤ 20 %	**2.1 %**	PASS
Average power	≤ 500 mW	—	needs device

Top-5 accuracy 88.8 % · sign model 5.8 MB INT8 (+13.6 MB MediaPipe) · power needs an on-phone reading

predict(video_path) → dict

```
{"clip_id": "17713",
 "prediction": "DRINK",
 "top5": ["DRINK","HOW",
   "FINISH","NOW","COLOR"],
 "confidence": 0.9878,
 "latency_ms": 3.02}
```

Deliverables

- ✓ predictions.jsonl (161 test clips)
- ✓ results.csv + benchmark_report.json
- ✓ signnet_int8.onnx + export.py
- ✓ Repo: predict(), README, Dockerfile
- ✓ Demo video (53 s, HUD + voice)

DIFFERENTIATION

Privacy by architecture

The classifier only sees 55 keypoints per frame. No faces, no backgrounds, no cloud.

Caption that respects ASL

The bubble floats beside the head and avoids the face and hands, which carry the grammar.

Honest uncertainty

Low-confidence guesses show in amber with a “?” instead of being stated as fact.

Robust by design

Mirror, yaw, roll and occlusion augmentation. Only −3 pt under low light, blur and occlusion.

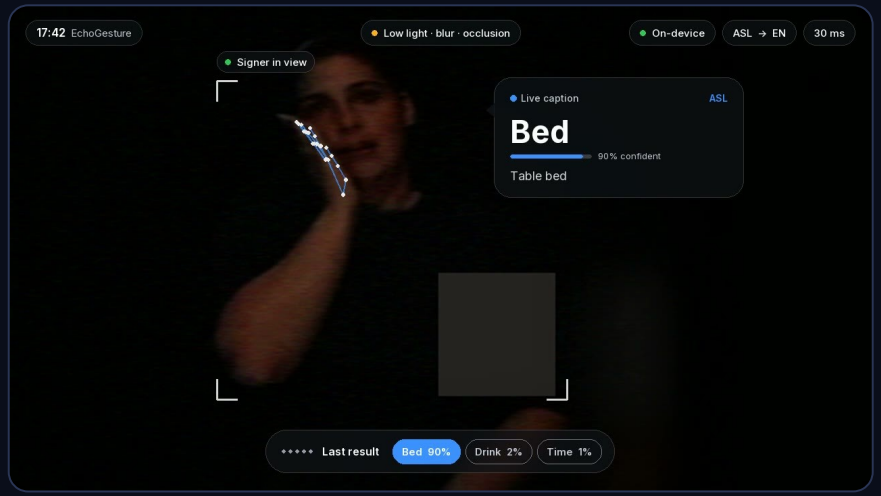
Fast where it counts

Parallel hand/pose trackers: p95 48.6 → 39.7 ms. Classifier 1.6 ms, flat after 10 min.

Evaluator-ready

One predict() call, Docker image, and one command that fills results.csv.

DEMO & LOCAL SETUP



Stress scene from the demo: low light · blur · occlusion

▸ Demo video (53 s)

[Demo video link: YouTube / Drive]

Run it locally

```
# setup (Python 3.11)
pip install -r requirements.txt

# evaluator entry point
python predict.py clip.mp4

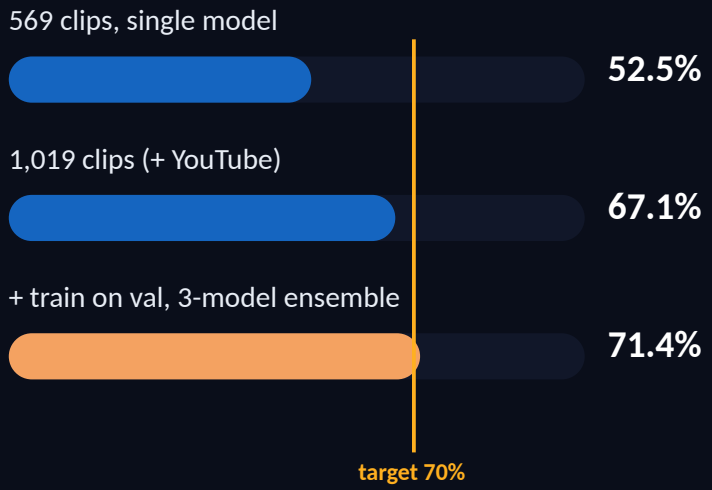
# whole test set -> predictions.jsonl
python scripts/make_predictions.py \
    --clips-dir test_clips/

# results.csv (+10-min thermal run)
python scripts/benchmark.py

# live AR caption HUD (webcam)
python demo/live_hud.py --tts
```

Brownie Slides

WHAT MOVED THE NEEDLE · test top-1



Limitations

- Only 1,019 of 2,038 WLASL100 clips were still online
- Isolated words, not continuous sentences
- Benchmarked on an M1 Mac, not a Galaxy phone yet
- Power needs an on-device reading

Next steps

- Android XR app: ONNX Runtime Mobile + MediaPipe Android
- Continuous signing with a CTC head
- Scale to WLASL2000 and How2Sign
- Speech → caption for the signer (two-way)

Thank You

EchoGesture · conversation without the gap, privacy without compromise